Reward Scaling Notes

Faraz Rahman

December 30, 2025

1 Intro

The goal of these notes are to provide some insight on generating / analyzing scaling curves of deep neural networks in environments (i.e. multistep tasks) where we are interested in "S"-shaped metrics like reward/value/returns which have upper and lower bounds.
Some common challanges in this setting are:

- Rewards vs. _____ follow an "S"-curve which is harder to paremetrize than e.g. $\log(\ell_{\text{val}})$ which is linear
- Each trial can be expensive to run, and various scales may require differing number of trials to acheive statistical certainty.

Agarwal et al. (2022) and Khatri et al. (2025) are papers that provide statistical tools / examples for benchmarking RL models in the few shot regime.

2 Sigmoidal Curves

References

Agarwal, R., Schwarzer, M., Castro, P. S., Courville, A., and Bellemare, M. G. (2022). Deep reinforcement learning at the edge of the statistical precipice.

Khatri, D., Madaan, L., Tiwari, R., Bansal, R., Duvvuri, S. S., Zaheer, M., Dhillon, I. S., Brandfonbrener, D., and Agarwal, R. (2025). The art of scaling reinforcement learning compute for llms.